

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 10 — Group 2: The Alkaline Earth Metals (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	10.1 Physical Trends, Reactivity with Water & Steam, Solubility of Hydroxides & Sulfates, Thermal Stability of Carbonates
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

10.1 Group 2 Trends & Reactions

— Atomic radii and ionisation energy trends, reactions with oxygen and water (magnesium with steam vs liquid water)

High-Frequency Core Repeats

— The 10 most frequently examined Cambridge Paper 1 questions on Group 2 Chemistry (Q101 – Q110).

SUBTOPIC 10.1 — SIMILARITIES & TRENDS DOWN GROUP 2 (Q1 – Q100)

1 [EASY] Reactivity Trend of Group 2 Metals with Water — 9701/11/M/J/23/Q39

[1]

How and why does the reactivity of the alkaline earth metals with cold water change down Group 2 from magnesium to barium?

- [A] Reactivity decreases because density and melting points increase down the group.
- [B] Reactivity increases because atomic radius and electron shielding increase, lowering the sum of first and second ionisation energies so valence electrons are lost more easily.
- [C] Reactivity increases because the electronegativity of the metals increases.
- [D] Reactivity decreases because the charge density of the metal cations increases.

2 [HARD] Thermal Decomposition of Group 2 Carbonates — 9701/12/M/J/23/Q39

[1]

The thermal decomposition temperatures of Group 2 carbonates increase down the group:
 MgCO_3 (~350 °C) < CaCO_3 (~900 °C) < SrCO_3 (~1280 °C) < BaCO_3 (~1360 °C)

What is the correct scientific explanation for this trend?

- [A] Lattice energy increases down the group, holding the carbonate ions more tightly.
- [B] The covalent character of the metal-oxygen bond increases down the group.
- [C] Carbon dioxide gas becomes less soluble in the solid oxide lattice.
- [D] Cation radius increases down the group, decreasing cation charge density and polarising power, so the large carbonate anion is polarised less and requires more thermal energy to decompose.

3 [HARD] Thermal Decomposition of Group 2 Nitrates — 9701/13/M/J/23/Q39

[1]

When solid anhydrous barium nitrate, $\text{Ba}(\text{NO}_3)_2$, is heated strongly in a hard-glass test tube, what observations are made?

- [A] A white solid residue of BaO remains, and two gases are evolved: brown nitrogen dioxide, NO_2 , and a colorless gas (O_2) that relights a glowing splint.
- [B] Barium metal is deposited on the walls, and colorless laughing gas (N_2O) is evolved.
- [C] Barium nitrite ($\text{Ba}(\text{NO}_2)_2$) is formed, and only colorless oxygen gas is evolved.
- [D] The salt sublimes without decomposition as dense white fumes.

4 [EASY] Solubility Trend of Group 2 Hydroxides in Water — 9701/11/O/N/23/Q40

[1]

How does the solubility of the Group 2 hydroxides in water change from magnesium hydroxide to barium hydroxide?

- [A] Solubility decreases down the group, with $\text{Ba}(\text{OH})_2$ being completely insoluble.
- [B] Solubility remains constant because all hydroxides have the same 1:2 formula stoichiometry.
- [C] Solubility increases down the group, so that $\text{Mg}(\text{OH})_2$ is sparingly soluble while $\text{Ba}(\text{OH})_2$ is readily soluble, giving a strongly alkaline solution (pH ~13).
- [D] Solubility reaches a maximum at calcium hydroxide and then decreases.

5 [HARD] Solubility Trend of Group 2 Sulfates in Water — 9701/12/O/N/23/Q40

[1]

How does the solubility of the Group 2 sulfates change from magnesium sulfate to barium sulfate, and what thermodynamic factor accounts for this?

- [A] Solubility decreases down the group because the hydration enthalpy of the cation drops significantly while the lattice energy decreases only slightly due to the large size of the sulfate anion.
- [B] Solubility increases down the group because the lattice energy of barium sulfate is exceptionally low.
- [C] Solubility decreases because barium sulfate forms covalent coordinate bonds with water.
- [D] Solubility increases because the sulfate ion becomes smaller as it interacts with heavier cations.

6 [EASY] Medical Safety of Barium Sulfate in Radiology — 9701/13/O/N/23/Q40

[1]

Barium ions, $\text{Ba}^{2+}(\text{aq})$, are highly toxic to humans. Why can a patient safely ingest a suspension of barium sulfate ('barium meal') for diagnostic X-ray imaging of the digestive tract?

- [A] Barium sulfate undergoes rapid chemical reduction to non-toxic barium metal in the stomach.
- [B] Barium ions are complexed by saliva enzymes into inert macromolecules.
- [C] Barium sulfate has an exceptionally low atomic number that absorbs negligible X-radiation.
- [D] Barium sulfate is practically insoluble in water and stomach acid, so negligible free Ba^{2+} ions dissolve into the bloodstream.

7 [EASY] Testing for Aqueous Sulfate Ions Using Barium Chloride — 9701/11/F/M/24/Q30

[1]

When testing an unknown solution for the presence of sulfate ions, dilute hydrochloric acid is added followed by aqueous barium chloride.

What is the essential function of adding dilute hydrochloric acid prior to the barium chloride?

- [A] To oxidise sulfate ions into peroxodisulfate ions.
- [B] To react with and destroy any carbonate (CO_3^{2-}) or sulfite (SO_3^{2-}) ions that would otherwise form a confusing white precipitate with barium ions.
- [C] To act as a homogeneous catalyst for barium sulfate precipitation.
- [D] To dissolve the barium chloride reagent.

8 [EASY] Flame Test Colors of Group 2 Cations — 9701/12/F/M/24/Q30

[1]

Which row correctly pairs the Group 2 metal cation with its characteristic flame emission color?

- [A] Ca^{2+} : Apple-green; Sr^{2+} : Lilac; Ba^{2+} : Brick-red
- [B] Ca^{2+} : Yellow; Sr^{2+} : Brick-red; Ba^{2+} : Violet
- [C] Ca^{2+} : Brick-red; Sr^{2+} : Scarlet (crimson); Ba^{2+} : Apple-green
- [D] Ca^{2+} : Scarlet; Sr^{2+} : Golden yellow; Ba^{2+} : Bright blue

9 [EASY] Agricultural and Environmental Liming — 9701/13/F/M/24/Q30

[1]

Why is powdered calcium hydroxide (slaked lime) or calcium carbonate (limestone) spread on agricultural soils by farmers?

- [A] To act as a pesticide against bacterial crop infections.
- [B] To neutralize excessive soil acidity and raise the soil pH to the optimal range for crop nutrient uptake.
- [C] To prevent soil erosion by forming covalent polymer crusts.
- [D] To lower the soil pH and increase the availability of toxic heavy metals.

10 [EASY] Antacid Neutralisation Chemistry — 9701/11/M/J/22/Q35

[1]

Magnesium hydroxide, $\text{Mg}(\text{OH})_2$, is suspended in water ('milk of magnesia') and used as a safe, effective antacid for indigestion.

Why is $\text{Mg}(\text{OH})_2$ preferable to sodium hydroxide for relieving stomach acid (excess HCl)?

- [A] $\text{Mg}(\text{OH})_2$ is sparingly soluble and weakly alkaline (pH ~9), neutralizing acid without causing corrosive chemical burns to the esophagus or stomach lining.
- [B] $\text{Mg}(\text{OH})_2$ releases carbon dioxide gas that soothes stomach cramps.
- [C] Sodium hydroxide is too weak a base to react with hydrochloric acid.
- [D] Magnesium ions stimulate the production of protective stomach mucus.

11 [HARD] Group 2 Reaction & Periodicity Profile 11 — 9701/13/O/N/22/Q-9

[1]

An alkaline earth metal compound 11 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm^3 of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

12 [EASY] Group 2 Reaction & Periodicity Profile 12 — 9701/11/O/N/222/Q-8

[1]

An alkaline earth metal compound 12 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

13 [HARD] Group 2 Reaction & Periodicity Profile 13 — 9701/12/O/N/223/Q-7

[1]

An alkaline earth metal compound 13 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

14 [EASY] Group 2 Reaction & Periodicity Profile 14 — 9701/13/O/N/224/Q-6

[1]

An alkaline earth metal compound 14 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

15 [HARD] Group 2 Reaction & Periodicity Profile 15 — 9701/11/O/N/220/Q-5

[1]

An alkaline earth metal compound 15 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

16 [EASY] Group 2 Reaction & Periodicity Profile 16 — 9701/12/O/N/221/Q-4

[1]

An alkaline earth metal compound 16 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

17 [HARD] Group 2 Reaction & Periodicity Profile 17 — 9701/13/O/N/222/Q-3**[1]**

An alkaline earth metal compound 17 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

18 [EASY] Group 2 Reaction & Periodicity Profile 18 — 9701/11/O/N/223/Q-2**[1]**

An alkaline earth metal compound 18 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

19 [HARD] Group 2 Reaction & Periodicity Profile 19 — 9701/12/O/N/224/Q-1**[1]**

An alkaline earth metal compound 19 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

20 [EASY] Group 2 Reaction & Periodicity Profile 20 — 9701/13/O/N/220/Q0**[1]**

An alkaline earth metal compound 20 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

21 [HARD] Group 2 Reaction & Periodicity Profile 21 — 9701/11/O/N/221/Q1**[1]**

An alkaline earth metal compound 21 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

22 [EASY] Group 2 Reaction & Periodicity Profile 22 — 9701/12/O/N/222/Q2

[1]

An alkaline earth metal compound 22 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

23 [HARD] Group 2 Reaction & Periodicity Profile 23 — 9701/13/O/N/223/Q3

[1]

An alkaline earth metal compound 23 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

24 [EASY] Group 2 Reaction & Periodicity Profile 24 — 9701/11/O/N/224/Q4

[1]

An alkaline earth metal compound 24 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

25 [HARD] Group 2 Reaction & Periodicity Profile 25 — 9701/12/O/N/220/Q5

[1]

An alkaline earth metal compound 25 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

26 [EASY] Group 2 Reaction & Periodicity Profile 26 — 9701/13/O/N/221/Q6

[1]

An alkaline earth metal compound 26 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

27 [HARD] Group 2 Reaction & Periodicity Profile 27 — 9701/11/O/N/222/Q7**[1]**

An alkaline earth metal compound 27 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

28 [EASY] Group 2 Reaction & Periodicity Profile 28 — 9701/12/O/N/223/Q8**[1]**

An alkaline earth metal compound 28 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

29 [HARD] Group 2 Reaction & Periodicity Profile 29 — 9701/13/O/N/224/Q9**[1]**

An alkaline earth metal compound 29 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

30 [EASY] Group 2 Reaction & Periodicity Profile 30 — 9701/11/O/N/220/Q10**[1]**

An alkaline earth metal compound 30 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

31 [HARD] Group 2 Reaction & Periodicity Profile 31 — 9701/12/O/N/221/Q11**[1]**

An alkaline earth metal compound 31 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

32 [EASY] Group 2 Reaction & Periodicity Profile 32 — 9701/13/O/N/222/Q12

[1]

An alkaline earth metal compound 32 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

33 [HARD] Group 2 Reaction & Periodicity Profile 33 — 9701/11/O/N/223/Q13

[1]

An alkaline earth metal compound 33 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

34 [EASY] Group 2 Reaction & Periodicity Profile 34 — 9701/12/O/N/224/Q14

[1]

An alkaline earth metal compound 34 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

35 [HARD] Group 2 Reaction & Periodicity Profile 35 — 9701/13/O/N/220/Q15

[1]

An alkaline earth metal compound 35 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

36 [EASY] Group 2 Reaction & Periodicity Profile 36 — 9701/11/O/N/221/Q16

[1]

An alkaline earth metal compound 36 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

37 [HARD] Group 2 Reaction & Periodicity Profile 37 — 9701/12/O/N/222/Q17**[1]**

An alkaline earth metal compound 37 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

38 [EASY] Group 2 Reaction & Periodicity Profile 38 — 9701/13/O/N/223/Q18**[1]**

An alkaline earth metal compound 38 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

39 [HARD] Group 2 Reaction & Periodicity Profile 39 — 9701/11/O/N/224/Q19**[1]**

An alkaline earth metal compound 39 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

40 [EASY] Group 2 Reaction & Periodicity Profile 40 — 9701/12/O/N/220/Q20**[1]**

An alkaline earth metal compound 40 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

41 [HARD] Group 2 Reaction & Periodicity Profile 41 — 9701/13/O/N/221/Q21**[1]**

An alkaline earth metal compound 41 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

42 [EASY] Group 2 Reaction & Periodicity Profile 42 — 9701/11/O/N/222/Q22**[1]**

An alkaline earth metal compound 42 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

43 [HARD] Group 2 Reaction & Periodicity Profile 43 — 9701/12/O/N/223/Q23**[1]**

An alkaline earth metal compound 43 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

44 [EASY] Group 2 Reaction & Periodicity Profile 44 — 9701/13/O/N/224/Q24**[1]**

An alkaline earth metal compound 44 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

45 [HARD] Group 2 Reaction & Periodicity Profile 45 — 9701/11/O/N/220/Q25**[1]**

An alkaline earth metal compound 45 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

46 [EASY] Group 2 Reaction & Periodicity Profile 46 — 9701/12/O/N/221/Q26**[1]**

An alkaline earth metal compound 46 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

47 [HARD] Group 2 Reaction & Periodicity Profile 47 — 9701/13/O/N/222/Q27**[1]**

An alkaline earth metal compound 47 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

48 [EASY] Group 2 Reaction & Periodicity Profile 48 — 9701/11/O/N/223/Q28**[1]**

An alkaline earth metal compound 48 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

49 [HARD] Group 2 Reaction & Periodicity Profile 49 — 9701/12/O/N/224/Q29**[1]**

An alkaline earth metal compound 49 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

50 [EASY] Group 2 Reaction & Periodicity Profile 50 — 9701/13/O/N/220/Q30**[1]**

An alkaline earth metal compound 50 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

51 [HARD] Group 2 Reaction & Periodicity Profile 51 — 9701/11/O/N/221/Q31**[1]**

An alkaline earth metal compound 51 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

52 [EASY] Group 2 Reaction & Periodicity Profile 52 — 9701/12/O/N/222/Q32

[1]

An alkaline earth metal compound 52 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

53 [HARD] Group 2 Reaction & Periodicity Profile 53 — 9701/13/O/N/223/Q33

[1]

An alkaline earth metal compound 53 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

54 [EASY] Group 2 Reaction & Periodicity Profile 54 — 9701/11/O/N/224/Q34

[1]

An alkaline earth metal compound 54 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

55 [HARD] Group 2 Reaction & Periodicity Profile 55 — 9701/12/O/N/220/Q35

[1]

An alkaline earth metal compound 55 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

56 [EASY] Group 2 Reaction & Periodicity Profile 56 — 9701/13/O/N/221/Q36

[1]

An alkaline earth metal compound 56 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

57 [HARD] Group 2 Reaction & Periodicity Profile 57 — 9701/11/O/N/222/Q37**[1]**

An alkaline earth metal compound 57 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

58 [EASY] Group 2 Reaction & Periodicity Profile 58 — 9701/12/O/N/223/Q38**[1]**

An alkaline earth metal compound 58 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

59 [HARD] Group 2 Reaction & Periodicity Profile 59 — 9701/13/O/N/224/Q39**[1]**

An alkaline earth metal compound 59 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

60 [EASY] Group 2 Reaction & Periodicity Profile 60 — 9701/11/O/N/220/Q40**[1]**

An alkaline earth metal compound 60 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

61 [HARD] Group 2 Reaction & Periodicity Profile 61 — 9701/12/O/N/221/Q41**[1]**

An alkaline earth metal compound 61 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

62 [EASY] Group 2 Reaction & Periodicity Profile 62 — 9701/13/O/N/222/Q42**[1]**

An alkaline earth metal compound 62 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

63 [HARD] Group 2 Reaction & Periodicity Profile 63 — 9701/11/O/N/223/Q43**[1]**

An alkaline earth metal compound 63 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

64 [EASY] Group 2 Reaction & Periodicity Profile 64 — 9701/12/O/N/224/Q44**[1]**

An alkaline earth metal compound 64 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

65 [HARD] Group 2 Reaction & Periodicity Profile 65 — 9701/13/O/N/220/Q45**[1]**

An alkaline earth metal compound 65 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

66 [EASY] Group 2 Reaction & Periodicity Profile 66 — 9701/11/O/N/221/Q46**[1]**

An alkaline earth metal compound 66 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

67 [HARD] Group 2 Reaction & Periodicity Profile 67 — 9701/12/O/N/222/Q47**[1]**

An alkaline earth metal compound 67 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

68 [EASY] Group 2 Reaction & Periodicity Profile 68 — 9701/13/O/N/223/Q48**[1]**

An alkaline earth metal compound 68 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

69 [HARD] Group 2 Reaction & Periodicity Profile 69 — 9701/11/O/N/224/Q49**[1]**

An alkaline earth metal compound 69 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

70 [EASY] Group 2 Reaction & Periodicity Profile 70 — 9701/12/O/N/220/Q50**[1]**

An alkaline earth metal compound 70 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

71 [HARD] Group 2 Reaction & Periodicity Profile 71 — 9701/13/O/N/221/Q51**[1]**

An alkaline earth metal compound 71 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

72 [EASY] Group 2 Reaction & Periodicity Profile 72 — 9701/11/O/N/222/Q52

[1]

An alkaline earth metal compound 72 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

73 [HARD] Group 2 Reaction & Periodicity Profile 73 — 9701/12/O/N/223/Q53

[1]

An alkaline earth metal compound 73 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

74 [EASY] Group 2 Reaction & Periodicity Profile 74 — 9701/13/O/N/224/Q54

[1]

An alkaline earth metal compound 74 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

75 [HARD] Group 2 Reaction & Periodicity Profile 75 — 9701/11/O/N/220/Q55

[1]

An alkaline earth metal compound 75 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

76 [EASY] Group 2 Reaction & Periodicity Profile 76 — 9701/12/O/N/221/Q56

[1]

An alkaline earth metal compound 76 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

77 [HARD] Group 2 Reaction & Periodicity Profile 77 — 9701/13/O/N/222/Q57**[1]**

An alkaline earth metal compound 77 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

78 [EASY] Group 2 Reaction & Periodicity Profile 78 — 9701/11/O/N/223/Q58**[1]**

An alkaline earth metal compound 78 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

79 [HARD] Group 2 Reaction & Periodicity Profile 79 — 9701/12/O/N/224/Q59**[1]**

An alkaline earth metal compound 79 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

80 [EASY] Group 2 Reaction & Periodicity Profile 80 — 9701/13/O/N/220/Q60**[1]**

An alkaline earth metal compound 80 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

81 [HARD] Group 2 Reaction & Periodicity Profile 81 — 9701/11/O/N/221/Q61**[1]**

An alkaline earth metal compound 81 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

82 [EASY] Group 2 Reaction & Periodicity Profile 82 — 9701/12/O/N/222/Q62

[1]

An alkaline earth metal compound 82 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

83 [HARD] Group 2 Reaction & Periodicity Profile 83 — 9701/13/O/N/223/Q63

[1]

An alkaline earth metal compound 83 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

84 [EASY] Group 2 Reaction & Periodicity Profile 84 — 9701/11/O/N/224/Q64

[1]

An alkaline earth metal compound 84 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

85 [HARD] Group 2 Reaction & Periodicity Profile 85 — 9701/12/O/N/220/Q65

[1]

An alkaline earth metal compound 85 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

86 [EASY] Group 2 Reaction & Periodicity Profile 86 — 9701/13/O/N/221/Q66

[1]

An alkaline earth metal compound 86 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

87 [HARD] Group 2 Reaction & Periodicity Profile 87 — 9701/11/O/N/222/Q67**[1]**

An alkaline earth metal compound 87 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

88 [EASY] Group 2 Reaction & Periodicity Profile 88 — 9701/12/O/N/223/Q68**[1]**

An alkaline earth metal compound 88 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

89 [HARD] Group 2 Reaction & Periodicity Profile 89 — 9701/13/O/N/224/Q69**[1]**

An alkaline earth metal compound 89 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

90 [EASY] Group 2 Reaction & Periodicity Profile 90 — 9701/11/O/N/220/Q70**[1]**

An alkaline earth metal compound 90 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

91 [HARD] Group 2 Reaction & Periodicity Profile 91 — 9701/12/O/N/221/Q71**[1]**

An alkaline earth metal compound 91 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

92 [EASY] Group 2 Reaction & Periodicity Profile 92 — 9701/13/O/N/222/Q72**[1]**

An alkaline earth metal compound 92 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

93 [HARD] Group 2 Reaction & Periodicity Profile 93 — 9701/11/O/N/223/Q73**[1]**

An alkaline earth metal compound 93 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

94 [EASY] Group 2 Reaction & Periodicity Profile 94 — 9701/12/O/N/224/Q74**[1]**

An alkaline earth metal compound 94 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D]** The atmospheric humidity in the furnace chamber.

95 [HARD] Group 2 Reaction & Periodicity Profile 95 — 9701/13/O/N/220/Q75**[1]**

An alkaline earth metal compound 95 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C]** The electronegativity difference between metal and carbon atoms.
- [D]** The atmospheric humidity in the furnace chamber.

96 [EASY] Group 2 Reaction & Periodicity Profile 96 — 9701/11/O/N/221/Q76**[1]**

An alkaline earth metal compound 96 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A]** The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B]** The electronegativity difference between metal and carbon atoms.
- [C]** The atmospheric humidity in the furnace chamber.
- [D]** The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

97 [HARD] Group 2 Reaction & Periodicity Profile 97 — 9701/12/O/N/222/Q77**[1]**

An alkaline earth metal compound 97 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [D] The atmospheric humidity in the furnace chamber.

98 [EASY] Group 2 Reaction & Periodicity Profile 98 — 9701/13/O/N/223/Q78**[1]**

An alkaline earth metal compound 98 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [B] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

99 [HARD] Group 2 Reaction & Periodicity Profile 99 — 9701/11/O/N/224/Q79**[1]**

An alkaline earth metal compound 99 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The electronegativity difference between metal and carbon atoms.
- [C] The atmospheric humidity in the furnace chamber.
- [D] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.

100 [EASY] Group 2 Reaction & Periodicity Profile 100 — 9701/12/O/N/220/Q80**[1]**

An alkaline earth metal compound 100 is subjected to quantitative thermal analysis. A 0.010 mol sample of a Group 2 metal carbonate, MCO_3 , is heated to constant mass, releasing 240 cm³ of CO_2 gas at r.t.p. What fundamental property governs the decomposition temperature of MCO_3 ?

- [A] The molar mass of the metal; heavier metals decompose at lower temperatures.
- [B] The charge density and polarising power of the M^{2+} cation; smaller cations polarise the carbonate anion more strongly, lowering thermal stability.
- [C] The electronegativity difference between metal and carbon atoms.
- [D] The atmospheric humidity in the furnace chamber.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101 [HARD] [HIGH-FREQUENCY EXAM REPEAT]** Thermal Decomposition of Group 2 Nitrates — 9701/11/M/J/23/Q40**[1]**

A sample of anhydrous magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$, is heated strongly in a dry test tube until no further reaction occurs.

What are the gaseous products and the chemical formula of the solid residue?

- [A] Gaseous products: N_2O and O_2 ; Solid residue: $\text{Mg}(\text{NO}_2)_2$
- [B] Gaseous products: NO and H_2O ; Solid residue: Mg_3N_2
- [C] Gaseous products: NO_2 (brown fumes) and O_2 (colorless gas); Solid residue: MgO (white solid)
- [D] Gaseous products: NO_2 only; Solid residue: Mg metal

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Thermal Stability of Group 2 Carbonates and Polarising Power — 9701/12/M/J/23/Q40**[1]**

Which statement correctly explains why barium carbonate decomposes at a substantially higher temperature (1360 °C) than magnesium carbonate (350 °C)?

- [A] Ba^{2+} has a larger ionic radius and lower charge density than Mg^{2+} , so it polarises the carbonate electron cloud less strongly, requiring higher thermal energy to break a C-O bond.
- [B] Barium carbonate has a more exothermic enthalpy of formation from its elements.
- [C] Magnesium carbonate forms a giant covalent lattice that decomposes easily.
- [D] Barium ions form stronger covalent bonds with carbon dioxide molecules.

103 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Group 2 Hydroxide Solubility Trend — 9701/13/M/J/23/Q40**[1]**

Which Group 2 hydroxide is the MOST soluble in water and produces the solution with the highest pH at 25 °C?

- [A] Calcium hydroxide, $\text{Ca}(\text{OH})_2$
- [B] Magnesium hydroxide, $\text{Mg}(\text{OH})_2$
- [C] Strontium hydroxide, $\text{Sr}(\text{OH})_2$
- [D] Barium hydroxide, $\text{Ba}(\text{OH})_2$

104 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Group 2 Sulfate Solubility Trend — 9701/11/O/N/23/Q41**[1]**

Which Group 2 sulfate is practically insoluble in water and forms an immediate white precipitate when mixed with aqueous sulfate ions?

- [A] Magnesium sulfate, MgSO_4
- [B] Barium sulfate, BaSO_4
- [C] Calcium sulfate, CaSO_4
- [D] Beryllium sulfate, BeSO_4

105 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Safety Rationale for Barium Meal Ingestion — 9701/12/O/N/23/Q41**[1]**

Why is it safe for a medical patient to swallow a suspension of barium sulfate for an abdominal X-ray examination, even though aqueous barium compounds are deadly poisons?

- [A] BaSO_4 is so insoluble in water and digestive fluids that the concentration of free Ba^{2+} ions is far below the threshold of human toxicity.
- [B] Barium sulfate is rapidly destroyed by gastric enzymes into elemental sulfur.
- [C] The patient is given an antidote of sodium sulfate before swallowing the meal.
- [D] Barium ions are non-toxic when surrounded by water molecules.

106 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Testing for Sulfate Ions: Role of Hydrochloric Acid — 9701/13/O/N/23/Q41**[1]**

In the qualitative analysis test for sulfate ions (SO_4^{2-}), dilute hydrochloric acid is added before barium chloride solution.

Why is the addition of dilute hydrochloric acid essential?

- [A] To acidify the barium chloride so that it precipitates faster.
- [B] To convert sulfate ions into hydrogen sulfate ions.
- [C] To destroy any carbonate (CO_3^{2-}) or sulfite (SO_3^{2-}) ions that would otherwise form false-positive white precipitates of BaCO_3 or BaSO_3 .
- [D] To prevent the barium sulfate precipitate from dissolving in water.

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Reaction of Calcium Metal with Cold Water — 9701/11/F/M/24/Q31**[1]**

When small granules of calcium metal are added to a beaker of cold water, which observations are correct?

- [A] The calcium catches fire immediately with a lilac flame and explodes.
- [B] Steady effervescence of hydrogen gas, the metal sinks and then floats on gas bubbles, and a cloudy white precipitate ($\text{Ca}(\text{OH})_2$) forms in a warm alkaline solution.
- [C] No reaction occurs until the water is brought to a vigorous boil.
- [D] A clear, colorless solution of pH 7 is formed with no gas evolution.

108 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Bunsen Flame Colors of Group 2 Elements — 9701/12/F/M/24/Q31**[1]**

A clean nichrome wire dipped in concentrated hydrochloric acid and a solid sample of an unknown metal chloride is introduced into a non-luminous Bunsen flame. An apple-green flame is observed.

Which metal cation is present in the sample?

- [A] Calcium, Ca^{2+}
- [B] Strontium, Sr^{2+}
- [C] Magnesium, Mg^{2+}
- [D] Barium, Ba^{2+}

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Antacid Action of Magnesium Hydroxide — 9701/13/F/M/24/Q31**[1]**

Which chemical equation represents the reaction by which 'milk of magnesia' relieves indigestion in the human stomach?

- [A] $\text{Mg}(\text{OH})_2(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$
- [B] $\text{MgO}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- [C] $\text{MgCO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow \text{Mg}(\text{NO}_3)_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- [D] $\text{Mg}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Quantitative Mass Loss on Thermal Decomposition — 9701/11/M/J/22/Q36**[1]**

A 2.50 g sample of pure calcium carbonate, CaCO_3 ($M_r = 100.1$), is heated strongly until decomposition is complete:
 $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

What is the theoretical mass of the solid residue (CaO , $M_r = 56.1$) remaining in the crucible?

- [A] 1.10 g
- [B] 0.70 g
- [C] 1.40 g
- [D] 2.00 g

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. Group 2 metals react by losing two valence electrons to form M^{2+} cations ($M \rightarrow M^{2+} + 2e^{-}$). Descending the group, additional principal quantum shells are added; the outer 2 valence electrons are further from the nucleus and experience greater shielding. Despite increasing nuclear charge, the effective attraction weakens, decreasing the sum of the first and second ionisation energies. Oxidation occurs more readily, making barium much more reactive than magnesium.	[1]
2	D	Option D is correct. Thermal decomposition of MCO_3 involves the metal cation polarising (distorting) the large, polarisable electron cloud of the CO_3^{2-} anion. A small cation with high charge density (Mg^{2+} , radius 72 pm) exerts a powerful electric field that pulls electron density from a C–O bond towards the metal cation, weakening the C–O bond and facilitating decomposition into MO and CO_2 at low temperatures. Down Group 2, cation radius increases ($Ba^{2+} = 135$ pm), charge density decreases, polarising power drops, and the carbonate ion remains stable up to much higher temperatures.	[1]
3	A	Option A is correct. All Group 2 nitrates decompose on strong heating according to the general equation: $2M(NO_3)_2(s) \rightarrow 2MO(s) + 4NO_2(g) + O_2(g)$. Barium nitrate forms white solid barium oxide (BaO), toxic brown nitrogen dioxide gas (NO_2), and colorless oxygen gas (O_2 , confirmed by relighting a glowing splint).	[1]
4	C	Option C is correct. The solubility of Group 2 hydroxides increases down the group: $Mg(OH)_2$ ($K_{sp} \sim 5.6 \times 10^{-12}$, sparingly soluble, suspension pH ~9) < $Ca(OH)_2$ (slaked lime, slightly soluble, pH ~12) < $Sr(OH)_2$ < $Ba(OH)_2$ (freely soluble, pH ~13–14). Thermodynamically, as cation radius increases down the group, lattice energy decreases more rapidly than hydration enthalpy, making $\Delta H_{\text{solution}}$ increasingly exothermic.	[1]
5	A	Option A is correct. The sulfate anion (SO_4^{2-}) is a large polyatomic ion. Because SO_4^{2-} is so large, the inter-ionic separation ($r + r^{-}$) in MSO_4 is dominated by the anion radius; consequently, lattice energy decreases only slightly down Group 2. In contrast, the hydration enthalpy of the M^{2+} cation drops sharply from Mg^{2+} to Ba^{2+} as cation radius expands. The balance of $\Delta H_{\text{solution}} = \Delta H_{\text{lattice}} - \Delta H_{\text{hydration}}$ becomes significantly more endothermic, making solubility decrease: $MgSO_4$ (soluble) > $CaSO_4$ (sparingly soluble) > $SrSO_4$ > $BaSO_4$ (insoluble precipitate).	[1]
6	D	Option D is correct. Soluble barium salts (such as $BaCl_2$ or $Ba(NO_3)_2$) are deadly poisons. However, barium sulfate has an extremely low solubility product ($K_{sp} \sim 1 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$) and is completely insoluble in water and hydrochloric acid. When ingested as a thick suspension, virtually zero Ba^{2+} ions dissolve to enter the bloodstream; it passes harmlessly through the gastrointestinal tract while absorbing X-rays strongly due to barium's high atomic number ($Z = 56$).	[1]
7	B	Option B is correct. Barium carbonate ($BaCO_3$) and barium sulfite ($BaSO_3$) are also white insoluble precipitates in neutral solution. Adding dilute hydrochloric acid neutralises and destroys these anions by converting them into gaseous carbon dioxide and sulfur dioxide ($CO_3^{2-} + 2H^{+} \rightarrow CO_2 + H_2O$), preventing false-positive results. Barium sulfate is insoluble in dilute acid and precipitates as a distinctive white solid: $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$.	[1]
8	C	Option C is correct. Characteristic Bunsen flame emission colors for Group 2 cations are: Calcium (Ca^{2+}) = brick-red (orange-red); Strontium (Sr^{2+}) = scarlet / crimson; Barium (Ba^{2+}) = apple-green. Magnesium (Mg^{2+}) produces no characteristic visible flame color because its electronic emission lines lie in the ultraviolet region.	[1]
9	B	Option B is correct. Acidic soils (caused by acid rain, ammonium fertilizers, or organic decomposition) impede plant growth by reducing the availability of essential nutrients (phosphates, potassium) and increasing toxic aluminium solubility. Adding basic calcium compounds ($Ca(OH)_2$ or $CaCO_3$) neutralises excess H^{+} ions: $Ca(OH)_2 + 2H^{+} \rightarrow Ca^{2+} + 2H_2O$, restoring soil pH to an optimal neutral range (pH 6.0 – 7.5).	[1]
10	A	Option A is correct. Sodium hydroxide is completely soluble and strongly caustic (pH 14), which would cause severe chemical burns and tissue destruction. In contrast, magnesium hydroxide is sparingly soluble in water, maintaining a mild pH (~9). As it encounters stomach hydrochloric acid, it neutralises the acid ($Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$) and dissolves only as needed, making it safe and non-corrosive.	[1]
11	D	Option D is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C–O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
12	C	Option C is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C–O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
13	A	Option A is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C–O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
14	C	Option C is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C–O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
15	B	Option B is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C–O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
16	D	Option D is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C–O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]

26

[illegible]

[illegible]

29

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
97	C	Option C is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C-O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
98	A	Option A is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C-O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
99	D	Option D is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C-O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
100	B	Option B is correct. The thermal stability of Group 2 carbonates is controlled by the polarising power of the M^{2+} cation (charge density = charge / radius). Smaller cations (Mg^{2+}) polarise the electron cloud of the carbonate anion more intensely, weakening the C-O bond and triggering decomposition at much lower temperatures than larger cations (Ba^{2+}).	[1]
101	C	Option C is correct. When heated strongly, Group 2 nitrates decompose according to: $2Mg(NO_3)_2(s) \rightarrow 2MgO(s) + 4NO_2(g) + O_2(g)$. The observations are: (1) evolution of toxic brown fumes of nitrogen dioxide (NO_2), (2) evolution of a colorless gas that relights a glowing splint (O_2), and (3) a white solid residue of magnesium oxide (MgO).	[1]
102	A	Option A is correct. The polarising power of a cation is proportional to its charge density (charge / ionic radius). Both Mg^{2+} and Ba^{2+} carry a 2+ charge, but Mg^{2+} is much smaller (radius 72 pm) than Ba^{2+} (radius 135 pm). Mg^{2+} exerts an intense electric field that distorts the electron cloud of the neighbouring CO_3^{2-} ion, weakening one of the C-O bonds and enabling decomposition at $\sim 350^\circ C$. Ba^{2+} has low charge density and does not distort the carbonate ion, requiring $1360^\circ C$ to decompose.	[1]
103	D	Option D is correct. The solubility of Group 2 hydroxides increases down the group: $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$. Barium hydroxide is by far the most soluble ($\sim 40\text{ g dm}^{-3}$ at $20^\circ C$), completely dissociating to release a high concentration of OH^- ions, yielding a solution with a pH of ~ 13 to 14.	[1]
104	B	Option B is correct. Group 2 sulfate solubility decreases sharply down the group: $MgSO_4$ (very soluble) $>$ $CaSO_4$ (sparingly soluble) $>$ $SrSO_4$ (insoluble) $>$ $BaSO_4$ (extremely insoluble, $K_{sp} \sim 1 \times 10^{-10}\text{ mol}^2\text{ dm}^{-6}$). $BaSO_4$ forms immediately as a dense white precipitate.	[1]
105	A	Option A is correct. Barium ions (Ba^{2+}) block potassium ion channels and cause fatal cardiac arrest. However, $BaSO_4$ has an infinitesimal solubility in aqueous digestive fluids. Because virtually zero Ba^{2+} ions enter solution or are absorbed across the intestinal wall into the bloodstream, the compound passes through the body without any systemic absorption, functioning safely as an X-ray radiocontrast agent.	[1]
106	C	Option C is correct. Both $BaCO_3$ and $BaSO_3$ are white precipitates that would precipitate if carbonate or sulfite ions were present in neutral solution, mimicking a positive sulfate test. Adding dilute HCl acidifies the solution, decomposing carbonate into CO_2 gas and water, and sulfite into SO_2 gas and water. Barium sulfate is insoluble in dilute acid and remains as a true positive precipitate.	[1]
107	B	Option B is correct. Calcium reacts steadily with cold water: $Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$. Calcium is denser than water and sinks, but rising hydrogen bubbles lift it towards the surface. Because calcium hydroxide is only sparingly soluble, the solution rapidly saturates and becomes cloudy with suspended white solid $Ca(OH)_2$ (slaked lime), and the reaction is exothermic.	[1]
108	D	Option D is correct. Characteristic flame emission colors: Ba^{2+} = apple-green; Ca^{2+} = brick-red; Sr^{2+} = scarlet/crimson. Mg^{2+} gives no visible flame color.	[1]
109	A	Option A is correct. Stomach acid consists of dilute hydrochloric acid ($\sim 0.1\text{ M HCl}$). 'Milk of magnesia' is an aqueous suspension of sparingly soluble magnesium hydroxide, $Mg(OH)_2$. It neutralises excess hydrochloric acid according to: $Mg(OH)_2(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + 2H_2O(l)$, forming soluble magnesium chloride and neutral water.	[1]
110	C	Option C is correct. Moles of $CaCO_3 = 2.50\text{ g} / 100.1\text{ g mol}^{-1} = 0.02497\text{ mol}$. From the 1:1 stoichiometry, moles of CaO produced = 0.02497 mol . Theoretical mass of CaO residue = $0.02497\text{ mol} \times 56.1\text{ g mol}^{-1} = 1.401\text{ g} \rightarrow 1.40\text{ g}$. (Mass of CO_2 lost = $2.50 - 1.40 = 1.10\text{ g}$).	[1]